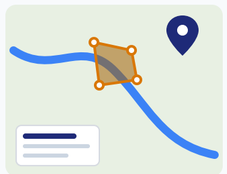
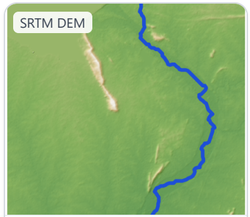
Hybrid physics: SPH where the flow is violent, 2D shallow water where the river is long.

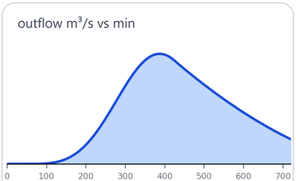
PIPELINE (images are real outputs of our engine - Teesta-III and Hirakud demo runs)



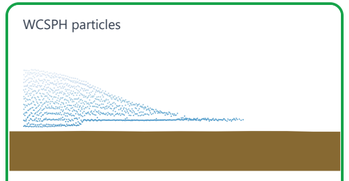
1 Input
Dam from register, custom pin, or **draw the landslide debris** on imagery



2 Pre-process
River trace, sill carving, lake volume curve

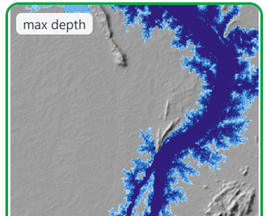


3 Breach hydrograph
Froehlich 2008 · MacDonald-LM · Costa & Schuster (landslide)

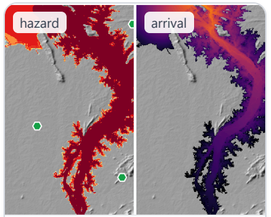


4a SPH near field
Breach + first 1-3 km · debris · **impact force** on barrage

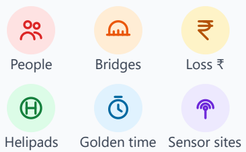
handover



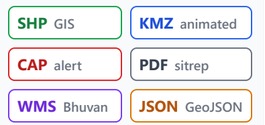
4b 2D far field
SWE-2D / Delft3D FM, 20-100 km, cascade dams



5 Hazard + timing
H1-H6 (depth × velocity), arrival time, probability map



6 Impact + HADR
Settlements, roads, hospitals, relief staging

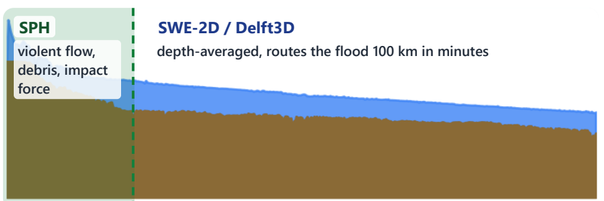


7 Export + Dashboard
Live map, 1-click sitrep, CAP for NDMA SACHET

TECH STACK

Solvers	SWE-2D (C#)	WCSPH (C#)	Delft3D FM	DualSPHysics GPU		
Geo	GLO-30 / SRTM	NetTopologySuite	OpenStreetMap	Earth Engine		
Backend	.NET 8	Clean Arch + CQRS	Hangfire	Redis	Neon Postgres	JWT
Frontend	React	MapLibre 3D	Recharts	SignalR		
Deploy	Docker Compose	K8s-ready	Offline sim core			

WHY HYBRID?



Breach peak (Froehlich 1995)

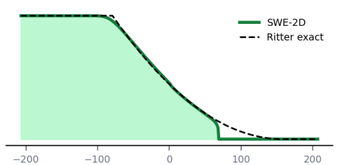
$$Q_p = 0.607\, V_w^{0.295}\, h_w^{1.24}$$

Overlap zone
SPH vs SWE vs Ritter compared at the same gauges

LAKE-TO-ALERT (AUTOMATIC)



VALIDATION LADDER



1	**Ritter / Stoker** exact solution	L1 2.9% / 0.8%
2	**Malpasset 1959** benchmark	runner ready
3	**Teesta 2023** vs Sentinel-1	CSI pending